Sniping Sequences

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Telescoping

$$\sum_{k=1}^{n} a_k - a_{k-1} = a_n - a_0$$

Basic excercises

1. $$\frac{1}{1\times 2} + \frac{1}{2\times 3} + \cdots + \frac{1}{9999\times 10000}$$

2. $$\sum_{k=0}^{2017} \frac{k}{(k+1)!}$$

3. $$\sum_{k=0}^{2017} \binom{k+8}{k}$$

4. $$\prod_{k=1}^{2017} \frac{k^2+5k+4}{k^2+5k+6}$$

5. Prove
$$1.5 < \sum_{n=1}^{\infty} \frac{1}{n^2} < 2$$

Solving recurrences

$$ax_{n+2} + bx_{n+1} + cx_n = 0 \implies x_n = A\alpha^n + B\beta^n$$

1. Find the general term of the fibonacci sequence $F_{n+2} = F_{n+1} + F_n$, $F_0 = 0, F_1 = 1$. Hence show that $F_{n+1}/F_n \to \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

2. Let the sequence a_n be defined as $a_{n+2} - ka_{n+1} + a_n = 0$, $a_0 = 2, a_1 = k$ for some real number $|k| < 2$. Prove that a_n is bounded.

Fun questions

1. $\mathbb{R}^*$ is the set of nonnegative reals. Find all $f : \mathbb{R}^* \to \mathbb{R}^*$ satisfying $f(x) \geq x$ and $f(f(x) - x) = 2x$.

2. (Balkan 2016) Find all injective functions $f : \mathbb{R} \to \mathbb{R}$ such that for every real number x and every positive integer n,
$$\left|\sum_{i=1}^{n} i\left(f(x+i+1) - f(f(x+i))\right)\right| < 2016$$

3. (AMO 1989) u_n satisfies $u_1 = 1$,
$$u_{n+1} = \frac{1}{u_1 + u_2 + \ldots u_n}$$
. Determine whether there exist m such that
$$\sum_{i=1}^{m} u_i \geq 1989$$

4. (SMMC 2017) Let $a_1, a_2, a_3 \ldots$ be the sequence of real numbers defined by $a_1 = 1$ and for $m \geq 2$
$$a_m = \frac{1}{a_1^2 + a_2^2 + \cdots + a_{m-1}^2}$$
Show whether there exist N such that
$$a_1 + a_2 + \cdots + a_N > 2017^{2017}$$

5. (ISL 2015) Suppose that a sequence $a_1, a_2, \ldots$ of positive real numbers satisfies
$$a_{k+1} \geq \frac{ka_k}{a_k^2 + (k-1)}$$
for every positive integer k. Prove that $a_1 + a_2 + \ldots + a_n \geq n$ for every $n \geq 2$.

6. (USAMTS) Determine the value of
$$S = \sum_{k=1}^{2016} \sqrt{1 + \frac{1}{k^2} + \frac{1}{(k+1)^2}}$$
.

7. (CMO 2017) The sequences $\{u_n\}$ and $\{v_n\}$ are defined by
$$u_0 = u_1 = 1$$
$$u_n = 2u_{n-1} - 3u_{n-2}, \quad (n \geq 2)$$
$$v_0 = a, v_1 = b, v_2 = c$$
$$v_n = v_{n-1} - 3v_{n-2} + 27v_{n-3}, \quad (n \geq 3)$$
. There exists a positive integer N such that when $n > N$, we have $u_n \mid v_n$. Prove that $3a = 2b + c$.

More questions

http://www.math.cmu.edu/∼lohp/docs/math/3-telescope-solns.pdf